
FAYSAL
CYBER SECURITY
PROTECT
DETECT
DEFEND
RESPOND
SECURE TODAY, SAFE TOMORROW

1.Create a directory named mid_exam with the following subdirectories: documents, scripts, logs, backup

```
┌──(kali㉿kali)-[~]
└─$ mkdir -p mid_exam/{documents,scripts,logs,beckup}

┌──(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  mid_exam  Music  Pictures  Public  shell.php  shell.txt  Templates  Videos

┌──(kali㉿kali)-[~]
└─$ cd mid_exam ; ls
beckup  documents  logs  scripts
```

2.Create a file named employee_info.txt inside the documents directory and add company information to it. Then copy the file to the backup directory and rename it as employee_backup.txt

```
┌──(kali㉿kali)-[~/mid_exam]
└─$ cd documents

┌──(kali㉿kali)-[~/mid_exam/documents]
└─$ vi employee_info.txt

┌──(kali㉿kali)-[~/mid_exam/documents]
└─$ cp -rf employee_info.txt ~/mid_exam/beckup

┌──(kali㉿kali)-[~/mid_exam/documents]
└─$ cd ~/mid_exam/beckup

┌──(kali㉿kali)-[~/mid_exam/beckup]
└─$ ls
employee_info.txt
```

3.Create a shell script named scan.sh inside the scripts directory containing a simple echo command. Give execute permission only to the owner and run the script.

```
┌──(kali㉿kali)-[~/mid_exam/scripts]
└─$ vi scan.sh

┌──(kali㉿kali)-[~/mid_exam/scripts]
└─$ chmod +x scan.sh

┌──(kali㉿kali)-[~/mid_exam/scripts]
└─$ ./scan.sh
Hello, World!
Welcome to Bash Scripting
```

4.Collect the following system information:IP address, Hostname, Logged-in user ,Ping test results

Save all outputs in a file named network_report.txt inside the logs directory

```
!#/bin/bash


    echo "Hostname:"
    hostname
    echo

    echo "Logged-in User:"
    whoami
    echo

    echo "IP Address:"
    hostname -I
    echo

    echo "Ping Test Results:"
    ping -c 4 google.com
    echo
~
~
~
```

```
┌──(kali㉿kali)-[~/mid_exam/logs]
└─$ vi  network_report.txt

┌──(kali㉿kali)-[~/mid_exam/logs]
└─$ chmod +x network_report.txt

┌──(kali㉿kali)-[~/mid_exam/logs]
└─$ ./network_report.txt
./network_report.txt: 1: !#/bin/bash: not found
Hostname:
kali

Logged-in User:
kali

IP Address:
192.168.78.128 172.17.0.1

Ping Test Results:
PING google.com (142.250.77.142) 56(84) bytes of data.
64 bytes from lcmaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=1 ttl=128 time=60.2 ms
64 bytes from lcmaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=2 ttl=128 time=91.6 ms
64 bytes from lcmaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=3 ttl=128 time=81.5 ms
64 bytes from lcmaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=4 ttl=128 time=59.6 ms

── google.com ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3089ms
rtt min/avg/max/mdev = 59.601/73.218/91.550/13.787 ms
```

5.Run a ping test on 8.8.8.8 and perform a basic Nmap scan on scanme.nmap.org. Save the results in a file named scan_result.txt inside the logs directory.Display currently running processes and monitor system activity using terminal-based utilities.



```
#!/bin/bash

echo " ping test on 8.8.8.8"
ping -c 4 8.8.8.8
echo

echo "namp on scanme.nmap.org"
nmap -sC -sV -O -T5 scanme.nmap.org
echo
```

```
┌──(kali㉿kali)-[~/mid_exam/logs]
└─$ vi scan_result.txt

┌──(kali㉿kali)-[~/mid_exam/logs]
└─$ ./scan_result.txt
 ping test on 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=56.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=67.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=67.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=62.7 ms

── 8.8.8.8 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3092ms
rtt min/avg/max/mdev = 56.416/63.561/67.933/4.577 ms

namp on scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 09:10 -0400
Stats: 0:01:13 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 98.60% done; ETC: 09:11 (0:00:00 remaining)
Stats: 0:01:13 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 98.60% done; ETC: 09:11 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.018s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 997 filtered tcp ports (no-response)
PORT     STATE SERVICE    VERSION
22/tcp   open  ssh        OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp   open  http       Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp open  nping-echo Nping echo
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: phone
Running (JUST GUESSING): Nokia Symbian OS (85%)
OS CPE: cpe:/o:nokia:symbian_os
Aggressive OS guesses: Nokia 3600i mobile phone (85%)
No exact OS matches for host (test conditions non-ideal).
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

6.Generate both MD5 and SHA256 hashes of the employee_info.txt file and save the outputs in a file named hash_report.txt inside the logs directory.

```
┌──(kali㉿kali)-[~/mid_exam/documents]
└─$ md5sum employee_info.txt >~/mid_exam/logs/hash_report.txt

┌──(kali㉿kali)-[~/mid_exam/documents]
└─$ sha256sum employee_info.txt >> ~/mid_exam/logs/hash_report.txt

┌──(kali㉿kali)-[~/mid_exam/documents]
└─$ cat ~/mid_exam/logs/hash_report.txt
b01af0cf97e519fc17c6bb36b3b5c2ec  employee_info.txt
4c5e4301c003dcee3260032ec924c0f8a912a6c17550b9aeab8c6673cd4fa835  employee_info.txt
```

7.Investigate authentication logs by displaying the last 15 lines and searching for the keyword "failed"

```
┌──(kali㉿kali)-[~/mid_exam/logs]
└─$ grep -i "failed" /var/log/auth.log > ~/mid_exam/logs/auth_failed_report.txt
grep: /var/log/auth.log: No such file or directory
```

8.Create a bash script named system_info.sh that displays:Current date, Current user, Current working directory, Hostname

```
┌──(kali㉿kali)-[~/mid_exam/documents]
└─$ vi system_info.sh

┌──(kali㉿kali)-[~/mid_exam/documents]
└─$ chmod +x system_info.sh

┌──(kali㉿kali)-[~/mid_exam/documents]
└─$ ./system_info.sh
Current Date:
Tue May 12 09:57:06 AM EDT 2026

Current User:
kali

Current Working Directory:
/home/kali/mid_exam/documents

Hostname:
kali

IP Address:
192.168.78.128 172.17.0.1

Ping Test Results:
PING google.com (142.250.205.142) 56(84) bytes of data.
64 bytes from pnmaaa-aw-in-f14.1e100.net (142.250.205.142): icmp_seq=1 ttl=128 time=71.4 ms
64 bytes from pnmaaa-aw-in-f14.1e100.net (142.250.205.142): icmp_seq=2 ttl=128 time=45.8 ms
64 bytes from pnmaaa-aw-in-f14.1e100.net (142.250.205.142): icmp_seq=3 ttl=128 time=50.2 ms
64 bytes from pnmaaa-aw-in-f14.1e100.net (142.250.205.142): icmp_seq=4 ttl=128 time=62.4 ms

— google.com ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3089ms
rtt min/avg/max/mdev = 45.824/57.467/71.409/10.094 ms
```